



Introduction to the Course



Video: Welcome to the Course
4 min



Video: Instructor Introductions
3 min



Reading: Syllabus
5 min



Reading: Required Hardware
5 min



Reading: Getting Help
2 min



Discussion Prompt: Meet and Greet
15 min

Introduction to Machine Learning

Embedded Machine Learning

Using Edge Impulse to Collect Data

Feature Extraction

Review



Syllabus

Please note that a large amount of foundational math goes into training neural networks and using them for inference. This course does not assume a background in such knowledge, and as such, it is meant as an introductory course in applying machine learning (using various tools and libraries) in embedded systems without needing to understand the finer details of neural networks (and other machine learning algorithms).

We will broadly cover the listed topics to get feel for how machine learning is used rather than spend time deeply covering the math behind it. The hope is to get you familiar enough with machine learning to start using and playing with it. We also hope to give you a starting vocabulary so that you know what to search for when you get stuck or give you some inspiration on new areas to dive deeper into.

There are other courses, books, and videos available that dive deep into the math behind machine learning. This course is an introduction to using some modern tools to get machine learning (specifically, neural networks) running on embedded devices, such as microcontrollers.

Prerequisites

You should be familiar with the following areas before starting this course:

- Algebra (you should know how to "solve for x" in mathematical equations)
- Arduino (if you plan to use the Arduino board for the demos and projects)

Please note that several advanced topics are discussed in the course, such as backpropagation, Fourier transform, filtering, etc. It helps if you are familiar with these topics, but they are not necessary for completing the course. They are mentioned to explain how things work and to give you some vocabulary if you wish to dive deeper into a particular subject area. However, quizzes and tests will not assume you have knowledge of these advanced areas.

Syllabus

Here is a broad outline of the topics that will be covered in the course: